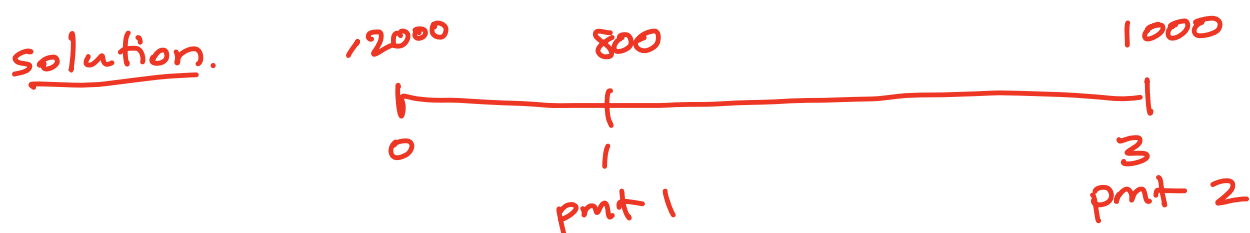


5.2 Amortized Loans and Schedules

definition. when a loan is an amortized loan interest due on the outstanding loan is first deducted from the payment amount.

The payment amount minus the interest due is called the principal

Problem: LaTroy took out a loan for \$2,000. The loan was governed by compound interest at a rate of 5%. He made a payment of \$800 one year later and a payment of \$1,000 three years after he took out the loan. Find the amount of principal in each payment and LaTroy's outstanding loan balance immediately after the second payment.



1st pmt: $2000(1.05) = 2100$ (100 of interest)
principal paid = $800 - 100 = 700$

Now DLB = 1300

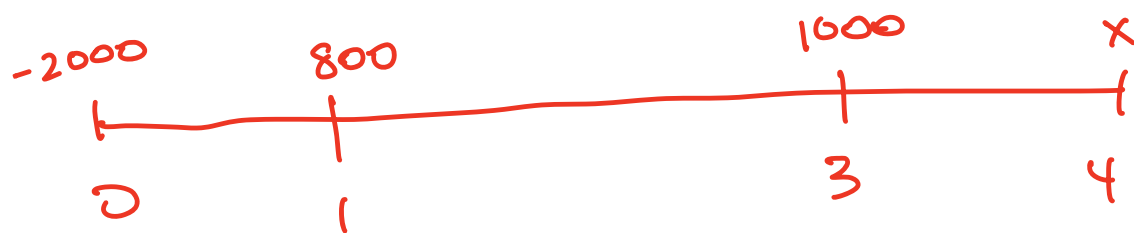
2nd pmt: $1300(1.05)^2 = 1433.25$ (133.25 of interest)
principal paid = $1000 - 133.25 = 866.75$
DLB = $1300 - 866.75 = 433.25$

Note the loans we calculated in Ch3 were amortized loans w/ level pmts at regular intervals.

For tax purposes its good to know the difference between interest pmts and principal pmts so we often create an

"Amortization Schedule"

lets consider the previous example if Latroy makes one final addition pmt at $t=4$



First note $2000(1.05)^4 = 800(1.05)^3 + 1000(1.05) + X$
so $X = 454.9125$ round to 454.91

PMT
Time

PMT

Loan
Balance

0	—	—	—	2000
1	800	100	700	1300
3	1000	133.25	866.75	433.25
4	454.91	21.66	433.25	0

Can we make an Amortization Table for a loan with rate i , n periods, pmts of Q (except final pmt) ?

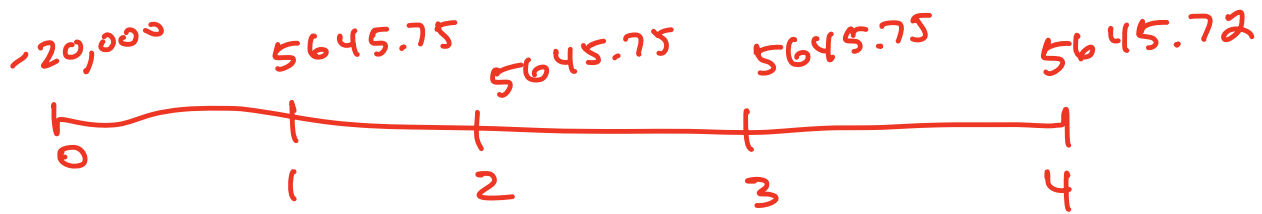
time	Pmt	Interest	Principal	OLB
0	—	—	—	$Qa_{\overline{n} i} = L$
1	Q	Li	$Q - Li$	$L - (Q - Li)$
2	Q			
3	Q			
⋮				
$n-1$	Q			
n	R			

Real example: $n = 60$, $i = .005$, $Q = 250$

$$\text{Loan amount} = 250 a_{\overline{60}|.05} = 12931.39019$$

Problem: Jared borrows \$20,000 at an annual effective discount rate of 4.8%. The loan is to be repaid by annual end-of-year payments for four years, the first three of which are for the amount \$5,645.75 and the time 4 payment is for \$5,645.72. Make an amortization schedule for this loan.

$$d = .048 \Rightarrow i = \frac{.048}{.952} = .050420168$$



t	PMT	Int.	Pri.	OLB
0	-	-	-	20,000
1	5645.75	1008.40	4637.35	15362.65
2	5645.75	774.59	4871.16	10491.49
3	5645.75	528.98	5116.77	5374.72
4	5645.72	271.00	5374.72	0

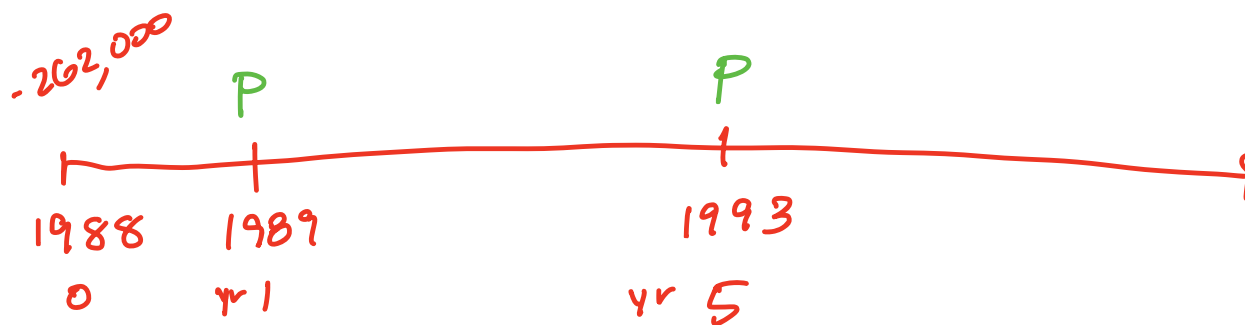
Problem: On May 1, 1988, the Ramakrishnas purchased a home for \$308,000. Their down payment was \$46,000, and the remaining \$262,000 was financed with a standard fifteen-year amortized loan at a nominal interest rate of 5.55% convertible monthly. How much interest did they pay in 1993, assuming that their payments have all been on the first of the month, the first payment was on June 1, 1988, and that there have been no additional payments beyond the scheduled ones?

Solution. $308,000 - 46,000 = 262,000$

$$15 \times 12 = 180 \text{ pmts}$$

$$i^{(12)} = .0555$$

$$i^{(12)}/12 = .004625$$



$$P a_{\overline{180}|.004625} = 262,000 \Rightarrow P = 2147.72$$

Amount of interest pmt $\underbrace{55 - 67}_{12 \text{ pmts}}$

$$= 10,922.22$$

5.4 Amortized Loans with other payment patterns

As long as the lender and borrower agree on the terms a loan can have unorthodox repayment schedules.

Nice Notation:

$L :=$ Loan Amount

$B_t :=$ Balance at time t (Immediately after any pmts at time t)

$P_t :=$ Pmt amount at time t (possibly 0)

$i_t :=$ Interest rate for $(t-1, t]$

Fact:
$$B_t = B_{t-1} (1 + i_t) - P_t$$

If n is the duration of the loan $B_n = 0$

Problem: Adam takes out a \$20,000 loan for ten years at a level annual effective interest rate of 5%. At the end of each year, he pays \$2,000 in principal, which is $\frac{1}{10}$ th of the loan amount, along with the interest due. Find a formula for P_t .

solution. $i = 5\%$ $L = 20,000$

end of 1st year : $20000(.05) = 1000$ ↖ interest
+ 2000 ↖ princ.
 $P_1 = 3000$

end of 2nd year $B_1 = 18,000$

$$: 18,000(.05) + 2000 = 2900$$

In general $B_t = 20,000 - 2000t$

so

$$20,000 - 2000t = (20000 - 2000(t-1))(1.05) - P_t$$

$$\begin{aligned} P_t &= 21,000 - 2100(t-1) - 20,000 + 2000t \\ &= 3100 - 100t \end{aligned}$$

Problem: Beatrice takes out a \$20,000 loan for ten years at a level annual effective interest rate of 5%. Her original payment is P and then her payments increase by \$200 each year. Find P .

Solution. If we know pmts are in Arith. Sequence we can leverage the P.V.

$$20,000 = (I_{P,200} a)_{\overline{10}|.05}$$

$$= P a_{\overline{10}|.05} + \frac{200}{.05} \left(a_{\overline{10}|.05} - \frac{10}{1.05^{10}} \right)$$

$$P = 1770.274 \rightarrow 1770.28$$

Problem: Cedric also takes out a \$20,000 loan for ten years at a level annual effective interest rate of 5%. His payments grow by 20% each year. Find the amount of his first payment.

solution.

$$20,000 = P \left(\frac{1 - \left(\frac{1.2}{1.05} \right)^{10}}{.05 - .2} \right) \Rightarrow P = 1070.97$$

Problem: Daphne also takes out a \$20,000 loan at a level annual effective interest rate of 5%. She makes annual payments that are four times the interest due until she can make a final end-of-year payment of no more than \$2,500 to pay off the loan. When does she make this final payment, and what is its amount?

$$P_1 = 20000 (.05)(4) = 4000$$

$$\begin{aligned} B_1 &= 20,000(1.05) - P_1 \\ &= 20,000(1.05) - 20,000(.05)(4) \\ &= 20,000(1 - 3(.05)) = 17000 \end{aligned}$$

$$P_2 = 20,000(1 - 3(.05))(.05)(4) = 3400$$

$$B_2 = 20,000(1 - 3(.05))(1.05) - P_2 = 14450$$

$$P_3 = 14450(.05)(4) = 2890$$

$$B_3 = 14450(1.05) - 2890 = 12282.5$$

$$B_t = (1.05)B_{t-1} - P_t \quad \text{and} \quad P_t = .2(B_{t-1})$$

$$\begin{aligned} B_t &= (1.05)B_{t-1} - .2(B_{t-1}) \\ &= .85B_{t-1} \end{aligned}$$

$$B_0 = 20,000, \quad B_1 = .85(20,000), \quad B_2 = .85^2(20,000)$$

$$B_t = 20,000(.85)^t$$

we want $B_t(1.05) < 2500$

$$20,000(.95)^t(1.05) < 2500$$

$$\Rightarrow t = 13.0952 \rightarrow t = 14$$

So in the 15th pmt: $B_{14}(1.05) = 2158.16$